File Organization

A database is mapped into a number of different files that are maintained by the underlying operating system. These files reside permanently on disks. A file is organized logically as a sequence of records. These records are mapped onto disk blocks. Files are provided as a basic construct in operating systems, so we shall assume the existence of an underlying file system. We need to consider ways of representing logical data models in terms of files.

Each file is also logically partitioned into fixed-length storage units called **blocks**, which are the units of both storage allocation and data transfer. Most databases use block sizes of 4 to 8 kilobytes by default, but many databases allow the block size to be specified when a database instance is created. Larger block sizes can be useful in some database applications.

A block may contain several records; the exact set of records that a block contains is determined by the form of physical data organization being used. We shall assume that no record is larger than a block. This assumption is realistic for most data-processing applications, such as our university example. There are certainly several kinds of large data items, such as images, that can be significantly larger than a block. Such large data items are stored separately, and a pointer to the data item is stored in the record.

In addition, we shall require that each record is entirely contained in a single block; that is, no record is contained partly in one block, and partly in another. This restriction simplifies and speeds up access to data items.

In a relational database, tuples of distinct relations are generally of different sizes. One approach to mapping the database to files is to use several files, and to store records of only one fixed length in any given file. An alternative is to structure our files so that we can accommodate multiple lengths for records; however, files of fixed-length records are easier to implement than are files of variable-length records. Many of the techniques used for the former can be applied to the variable-length case. Thus, we begin by considering a file of fixed-length records, and consider storage of variable-length records later.

Fixed-Length Record

As an example, let us consider a file of instructor records for our university database. Each record of this file is defined (in pseudocode) as:

```
type instructor = record
                      ID varchar (5);
                      name varchar(20);
                      dept_name varchar (20);
                      salary numeric (8,2);
                  end
```

Assume that each character occupies 1 byte and that numeric (8,2) occupies 8 bytes. Suppose that instead of allocating a variable amount of bytes for the attributes ID, name, and dept_name, we allocate the maximum number of bytes that each attribute can hold. Then, the instructor record is 53 bytes long. A simple approach is to use the first 53 bytes for the first record, the next 53 bytes for the second record, and so on (Figure 10.4). However, there are two problems with this simple approach:

record 0	10101	Srinivasan	Comp. Sci.	65000
record 1	12121	Wu	Finance	90000
record 2	15151	Mozart	Music	40000
record 3	22222	Einstein	Physics	95000
record 4	32343	El Said	History	60000
record 5	33456	Gold	Physics	87000
record 6	45565	Katz	Comp. Sci.	75000
record 7	58583	Califieri	History	62000
record 8	76543	Singh	Finance	80000
record 9	76766	Crick	Biology	72000
record 10	83821	Brandt	Comp. Sci.	92000
record 11	98345	Kim	Elec. Eng.	80000

Figure 10.4 File containing *instructor* records.

1. Unless the block size happens to be a multiple of 53 (which is unlikely), some records will cross block boundaries. That is, part of the record will be stored in

one block and part in another. It would thus require two block accesses to read or write such a record.

2. It is difficult to delete a record from this structure. The space occupied by the record to be deleted must be filled with some other record of the file, or we must have a way of marking deleted records so that they can be ignored.

To avoid the first problem, we allocate only as many records to a block as would fit entirely in the block (this number can be computed easily by dividing the block size by the record size, and discarding the fractional part). Any remaining bytes of each block are left unused.

When a record is deleted, we could move the record that came after it into the space formerly occupied by the deleted record, and so on, until every record following the deleted record has been moved ahead (Figure 10.5).

record 0	10101	Srinivasan	Comp. Sci.	65000
record 1	12121	Wu	Finance	90000
record 2	15151	Mozart	Music	40000
record 4	32343	El Said	History	60000
record 5	33456	Gold	Physics	87000
record 6	45565	Katz	Comp. Sci.	75000
record 7	58583	Califieri	History	62000
record 8	76543	Singh	Finance	80000
record 9	76766	Crick	Biology	72000
record 10	83821	Brandt	Comp. Sci.	92000
record 11	98345	Kim	Elec. Eng.	80000

Figure 10.5 File of Figure 10.4, with record 3 deleted and all records moved.

Such an approach requires moving a large number of records. It might be easier simply to move the final record of the file into the space occupied by the deleted record (Figure 10.6). It is undesirable to move records to occupy the space freed by a deleted record, since doing so requires additional block accesses. Since insertions tend to be more frequent than deletions, it is acceptable to leave open the space occupied by the deleted record, and to wait for a subsequent insertion before reusing the space. A simple marker on a deleted record is not sufficient, since it is hard to find this available space when an insertion is being done. Thus, we need to introduce an additional structure.

record 0	10101	Srinivasan	Comp. Sci.	65000
record 1	12121	Wu	Finance	90000
record 2	15151	Mozart	Music	40000
record 11	98345	Kim	Elec. Eng.	80000
record 4	32343	El Said	History	60000
record 5	33456	Gold	Physics	87000
record 6	45565	Katz	Comp. Sci.	75000
record 7	58583	Califieri	History	62000
record 8	76543	Singh	Finance	80000
record 9	76766	Crick	Biology	72000
record 10	83821	Brandt	Comp. Sci.	92000

Figure 10.6 File of Figure 10.4, with record 3 deleted and final record moved.

At the beginning of the file, we allocate a certain number of bytes as a **file header**. The header will contain a variety of information about the file. For now, all we need to store there is the address of the first record whose contents are deleted. We use this first record to store the address of the second available record, and so on. Intuitively, we can think of these stored addresses as pointers, since they point to the location of a record. The deleted records thus form a linked list, which is often referred to as a **free list**. Figure 10.7 shows the file of Figure 10.4, with the free list, after records 1, 4, and 6 have been deleted.

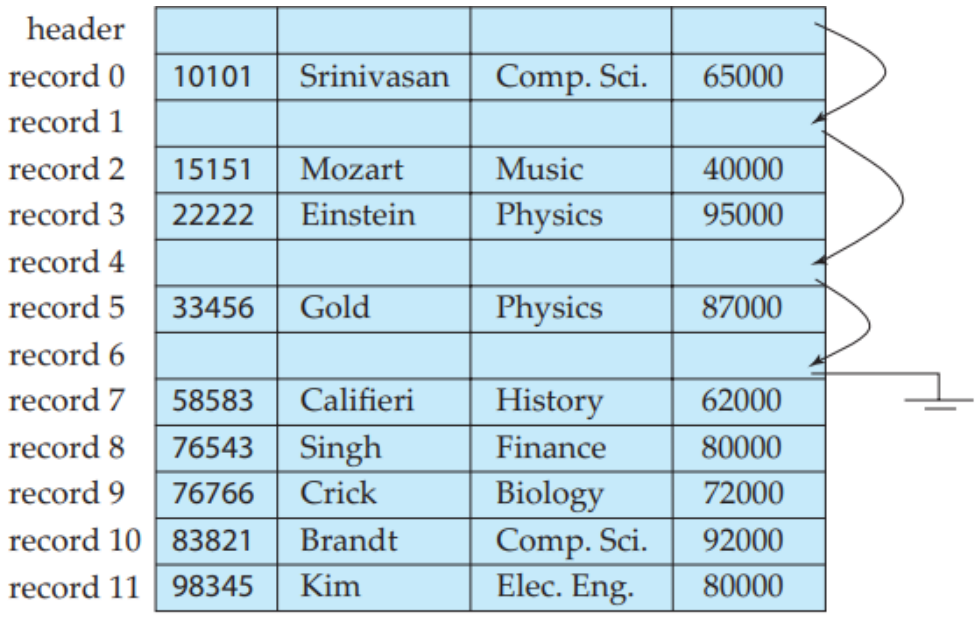


Figure 10.7 File of Figure 10.4, with free list after deletion of records 1, 4, and 6.

On insertion of a new record, we use the record pointed to by the header. We change the header pointer to point to the next available record. If no space is available, we add the new record to the end of the file.
Insertion and deletion for files of fixed-length records are simple to implement, because the space made available by a deleted record is exactly the space needed to insert a record. If we allow records of variable length in a file, this match no longer holds. An inserted record may not fit in the space left free by a deleted record, or it may fill only part of that space.

Variable-Length Records

Variable-length records arise in database systems in several ways:

- Storage of multiple record types in a file.
- Record types that allow variable lengths for one or more fields.
- Record types that allow repeating fields, such as arrays or multisets.

Different techniques for implementing variable-length records exist. Two different problems must be solved by any such technique:

- How to represent a single record in such a way that individual attributes can be extracted easily.
- How to store variable-length records within a block, such that records in a block can be extracted easily.

The representation of a record with variable-length attributes typically has two parts: an initial part with *fixed length* attributes, followed by data for *variable length* attributes. Fixed-length attributes, such as numeric values, dates, or fixed length character strings are allocated as many bytes as required to store their value. Variable-length attributes, such as varchar types, are represented in the initial part of the record by a pair (*offset, length*), where offset denotes where the data for that attribute begins within the record, and length is the length in bytes of the variable-sized attribute. The values for these attributes are stored consecutively, after the initial fixed-length part of the record. Thus, the initial part of the record stores a fixed size of information about each attribute, whether it is fixed-length or variable-length.
An example of such a record representation is shown in Figure 10.8. The figure shows an instructor record, whose first three attributes ID, name, and dept_name are variable-length strings, and whose fourth attribute salary is a fixed-sized number. We assume that the offset and length values are stored in two bytes each, for a total of 4

bytes per attribute. The salary attribute is assumed to be stored in 8 bytes, and each string takes as many bytes as it has characters.

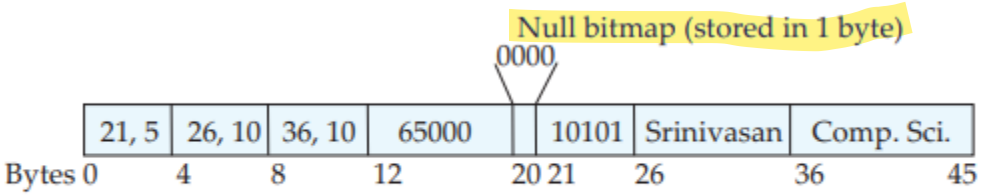


Figure 10.8 Representation of variable-length record.

The figure also illustrates the use of a null bitmap, which indicates which attributes of the record have a null value. In this particular record, if the salary were null, the fourth bit of the bitmap would be set to 1, and the salary value stored in bytes 12 through 19 would be ignored. Since the record has four attributes, the null bitmap for this record fits in 1 byte, although more bytes may be required with more attributes. In some representations, the null bitmap is stored at the beginning of the record, and for attributes that are null, no data (value, or offset/length) are stored at all. Such a representation would save some storage space, at the cost of extra work to extract attributes of the record. This representation is particularly useful for certain applications where records have a large number of fields, most of which are null.

We next address the problem of storing variable-length records in a block. The **slotted-page structure** is commonly used for organizing records within a block, and is shown in Figure 10.9.

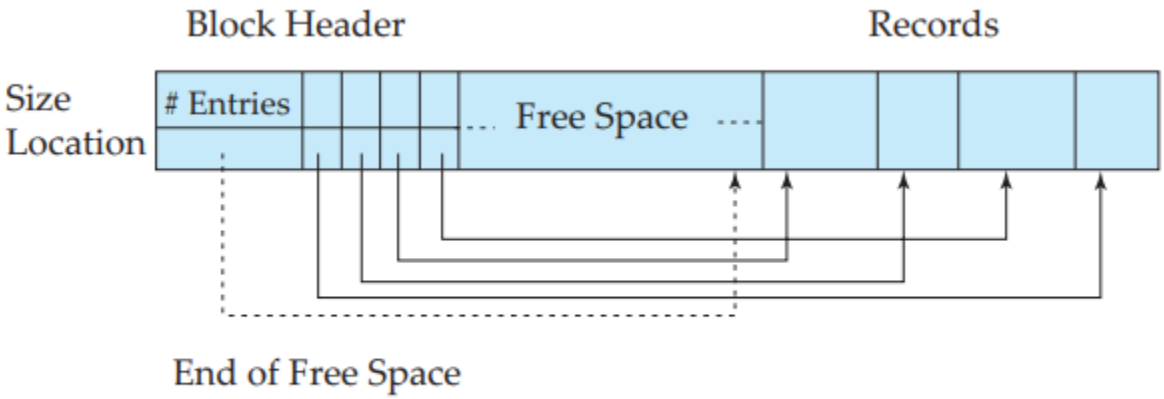


Figure 10.9 Slotted-page structure.

There is a header at the beginning of each block, containing the following information:

1. The number of record entries in the header.
2. The end of free space in the block.
3. An array whose entries contain the location and size of each record.

The actual records are allocated contiguously in the block, starting from the end of the block. The free space in the block is contiguous, between the final entry in the header array, and the first record. If a record is inserted, space is allocated for it at the end of free space, and an entry containing its size and location is added to the header.

If a record is deleted, the space that it occupies is freed, and its entry is set to deleted (its size is set to −1, for example). Further, the records in the block before the deleted record are moved, so that the free space created by the deletion gets occupied, and all free space is again between the final entry in the header array and the first record. The end-of-free-space pointer in the header is appropriately updated as well. Records can be grown or shrunk by similar techniques, as long as there is space in the block. The cost of moving the records is not too high, since the size of a block is limited: typical values are around 4 to 8 kilobytes.

The slotted-page structure requires that there be no pointers that point directly to records. Instead, pointers must point to the entry in the header that contains the actual location of the record. This level of indirection allows records to be moved to prevent fragmentation of space inside a block, while supporting indirect pointers to the record.

Storing large objects

Databases often store data that can be much larger than a disk block. For instance, an image or an audio recording may be multiple megabytes in size, while a video object may be multiple gigabytes in size. Recall that SQL supports the types **blob** and **clob**, which store binary and character large objects.
Most relational databases restrict the size of a record to be no larger than the size of a block, to simplify buffer management and free-space management.
Large objects are often stored in a special file (or collection of files) instead of being stored with the other (short) attributes of records in which they occur. A (logical) pointer to the object is then stored in the record containing the large object.
However, there are some performance issues with storing these large objects in databases. The efficiency of accessing large objects via database interfaces is one concern. A second concern is the size of database backups.

Organization of Records in Files

So far, we have studied how records are represented in a file structure. A relation is a set of records. Given a set of records, the next question is how to organize them in a file. Several of the possible ways of organizing records in files are:

Heap file organization
Any record can be placed anywhere in the file where there is space for the record. There is no ordering of records. Typically, there is a single file for each relation.

Sequential file organization
Records are stored in sequential order, according to the value of a "search key" of each record.

Hashing file organization
A hash function is computed on some attribute of each record. The result of the hash function specifies in which block of the file the record should be placed.

Multitable clustering File Organization
Generally, a separate file or set of files is used to store the records of each relation. However, in a multitable clustering file organization, records of several different relations are stored in the same file, and in fact in the same block within a file, to reduce the cost of certain join operations.

B+ tree file organization
The traditional sequential file organization does support ordered access even if there are insert, delete, and update operations, which may change the ordering of records. However, in the face of a large number of such operations, efficiency of ordered access suffers. We study another way of organizing records, called the B+-tree file organization.

Heap File Organization

In a heap file organization, a record may be stored anywhere in the file corresponding to a relation. Once placed in a particular location, the record is not usually moved. When a record is inserted, store the record at the end of the file. However, if a record gets deleted, it can be used to store new records.

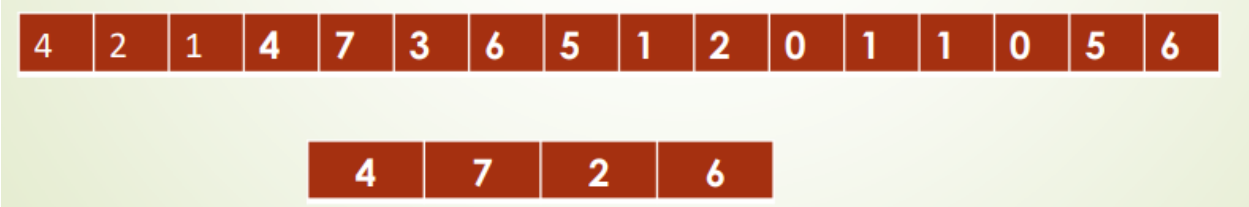
Most database structures use a space efficient data structure called free-space map to track which blocks have free space to store records. The free space map is commonly represented by an array containing 1 entry for each block in the relation. Each entry represents a fraction f such that at least a fraction f of the space in the block is free.

An example of a free-space map for a file with 16 blocks is shown below. We assume that 3 bits are used to store the occupancy fraction; the value at position i should be divided by 8 to get the free-space fraction for block i.

In PostgreSQL, an entry is 1 byte. So the entry is divided by 256.
To find a block to store a new record of a given size, the database can scan the freespace map to find a block that has enough free space to store that record. If there is no such block, a new block is allocated for the relation.

While such a scan is much faster than actually fetching blocks to find free space, it can still be very slow for large files. To further speed up the task of locating a block with sufficient free space, we can create a second-level free-space map, which has, say 1 entry for every 100 entries of the main free-spacemap. That 1 entry stores the maximum value amongst the 100 entries in the main free-space map that it corresponds to.

The free-space map below is a second level free-space map for our earlier example, with 1 entry for every 4 entries in the main free-space map. To deal with very large relations, we can create more levels beyond the second level, using the same idea.



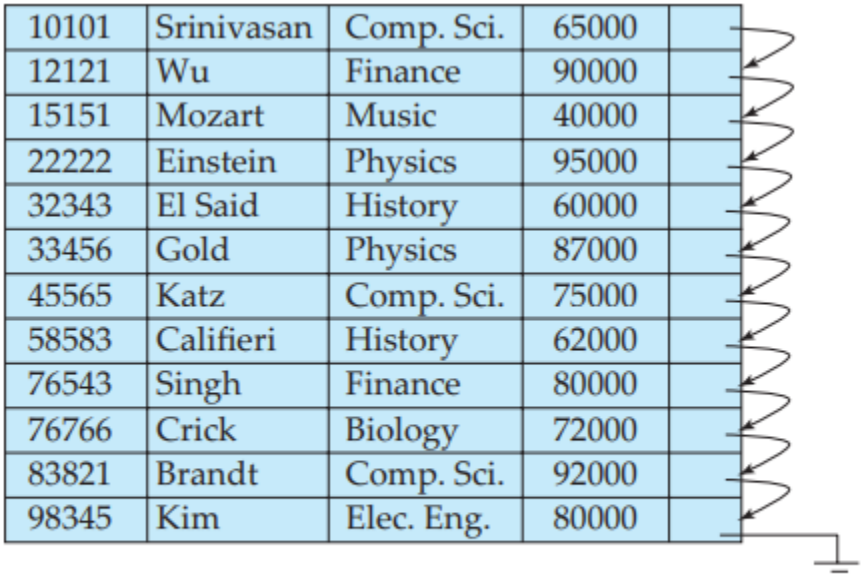
Demerits : Writing the free-space map to disk every time an entry in the map is updated would be very expensive. Instead, the free-space map is written periodically; as a result, the free-space map on disk may be outdated, and when a database starts up, it may get outdated data about available free space.

The free-space map may, as a result, claim a block has free space when it does not; such an error will be detected when the block is fetched, and can be dealt with by a further search in the free-space map to find another block.

On the other hand, the free-space map may claim that a block does not have free space when it does; generally this will not result in any problem other than unused free space. To fix any such errors, the relation is scanned periodically and the free-space map recomputed and written to disk.

Sequential File Organization

A sequential file is designed for efficient processing of records in sorted order based on some search key. A search key is any attribute or set of attributes; it need not be the primary key, or even a superkey. To permit fast retrieval of records in search-key order, we chain together records by pointers. The pointer in each record points to the next record in search-key order. Furthermore, to minimize the number of block accesses in sequential file processing, we store records physically in search-key order, or as close to search-key order as possible.



10101	Srinivasan	Comp. Sci.	65000	
12121	Wu	Finance	90000	
15151	Mozart	Music	40000	
22222	Einstein	Physics	95000	
32343	El Said	History	60000	
33456	Gold	Physics	87000	
45565	Katz	Comp. Sci.	75000	
58583	Califieri	History	62000	
76543	Singh	Finance	80000	
76766	Crick	Biology	72000	
83821	Brandt	Comp. Sci.	92000	
98345	Kim	Elec. Eng.	80000	

Figure 10.10 Sequential file for *instructor* records.

Figure 10.10 shows a sequential file of instructor records taken from our university example. In that example, the records are stored in search-key order, using ID as the search key.

The sequential file organization allows records to be read in sorted order; that can be useful for display purposes as well as for certain query-processing algorithms. It is difficult, however, to maintain physical sequential order as records are inserted and deleted, since it is costly to move many records as a result of a single insertion or deletion. We can manage deletion by using pointer chains, as we saw previously. For insertion, we apply the following rules:

1. Locate the record in the file that comes before the record to be inserted in search-key order.
2. If there is a free record (that is, space left after a deletion) within the same block as this record, insert the new record there. Otherwise, insert the new record in an overflow block. In either case, adjust the pointers so as to chain together the records in search-key order.

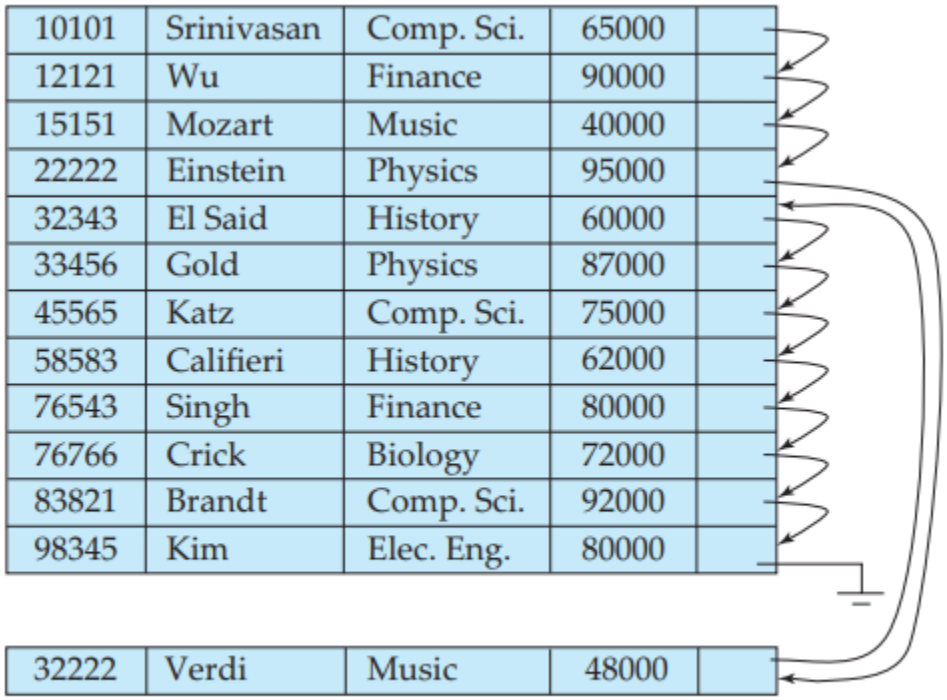


Figure 10.11 Sequential file after an insertion.

Figure 10.11 shows the file of Figure 10.10 after the insertion of the record (32222, Verdi, Music, 48000). The structure in Figure 10.11 allows fast insertion of new records, but forces sequential file-processing applications to process records in an order that does not match the physical order of the records.

If relatively few records need to be stored in overflow blocks, this approach works well. Eventually, however, the correspondence between search-key order and physical order may be totally lost over a period of time, in which case sequential processing will become much less efficient. At this point, the file should be **reorganized** so that it is once again physically in sequential order. Such reorganizations are costly, and must be done during times when the system load is low. The frequency with which reorganizations are needed depends on the frequency of insertion of new records. In the extreme case in which insertions rarely occur, it is possible always to keep the file in physically sorted order. In such a case, the pointer field in Figure 10.10 is not needed.

Multitable Clustering File Organization

Many relational database systems store each relation in a separate file, so that they can take full advantage of the file system that the operating system provides. Usually, tuples of a relation can be represented as fixed-length records. Thus, relations can be mapped to a simple file structure. This simple implementation of a relational database system is well suited to low-cost database implementations as in, for example, embedded systems or portable devices. In such systems, the size of the database is small, so little is gained from a sophisticated file structure. Furthermore, in such environments, it is essential that the overall size of the object code for the database system be small. A simple file structure reduces the amount of code needed to implement the system.

This simple approach to relational database implementation becomes less satisfactory as the size of the database increases. We have seen that there are performance advantages to be gained from careful assignment of records to blocks, and from careful organization of the blocks themselves. Clearly, a more complicated file structure may be beneficial, even if we retain the strategy of storing each relation in a separate file.

However, many large-scale database systems do not rely directly on the underlying operating system for file management. Instead, one large operating system file is allocated to the database system. The database system stores all relations in this one file, and manages the file itself.

Even if multiple relations are stored in a single file, by default most databases store records of only one relation in a given block. This simplifies data management. However, in some cases it can be useful to store records of more than one relation in a single block. To see the advantage of storing records of multiple relations in one block, consider the following SQL query for the university database:

```
select dept_name, building, budget, ID, name, salary
from department natural join instructor;
```

This query computes a join of the department and instructor relations. Thus, for each tuple of department, the system must locate the instructor tuples with the same value for dept_name. Ideally, these records will be located with the help of indices. Regardless of how these records are located, however, they need to be transferred from disk into main memory. In the worst case, each record will reside on a different block, forcing us to do one block read for each record required by the query.

As a concrete example, consider the department and instructor relations of Figures 10.12 and 10.13, respectively. In Figure 10.14, we show a file structure designed for efficient execution of queries involving the natural join of department and instructor. The instructor tuples for each ID are stored near the department tuple for the corresponding dept_name. This structure mixes together tuples of two relations, but allows for efficient processing of the join. When a tuple of the department relation is read, the entire block containing that tuple is copied from disk into main memory. Since the corresponding instructor tuples are stored on the disk near the department tuple, the block containing the department tuple contains tuples of the instructor relation needed to process the query. If a department has so many instructors that the instructor records do not fit in one block, the remaining records appear on nearby blocks.

dept_name	*building*	*budget*
Comp. Sci.	Taylor	100000
Physics	Watson	70000

Figure 10.12 The *department* relation.

ID	*name*	*dept_name*	*salary*
10101	Srinivasan	Comp. Sci.	65000
33456	Gold	Physics	87000
45565	Katz	Comp. Sci.	75000
83821	Brandt	Comp. Sci.	92000

Figure 10.13 The *instructor* relation.

Comp. Sci.	Taylor	100000
45564	Katz	75000
10101	Srinivasan	65000
83821	Brandt	92000
Physics	Watson	70000
33456	Gold	87000

Figure 10.14 Multitable clustering file structure.

A multitable clustering file organization is a file organization, such as that illustrated in Figure 10.14, that stores related records of two or more relations in each block. Such a

file organization allows us to read records that would satisfy the join condition by using one block read. Thus, we are able to process this particular query more efficiently.

In the representation shown in Figure 10.14, the dept_name attribute is omitted from instructor records since it can be inferred from the associated department record; the attribute may be retained in some implementations, to simplify access to the attributes. We assume that each record contains the identifier of the relation to which it belongs, although this is not shown in Figure 10.14.

Our use of clustering of multiple tables into a single file has enhanced processing of a particular join (that of department and instructor), but it results in slowing processing of other types of queries. For example, select * from department; requires more block accesses than it did in the scheme under which we stored each relation in a separate file, since each block now contains significantly fewer department records. To locate efficiently all tuples of the department relation in the structure of Figure 10.14, we can chain together all the records of that relation using pointers, as in Figure 10.15.

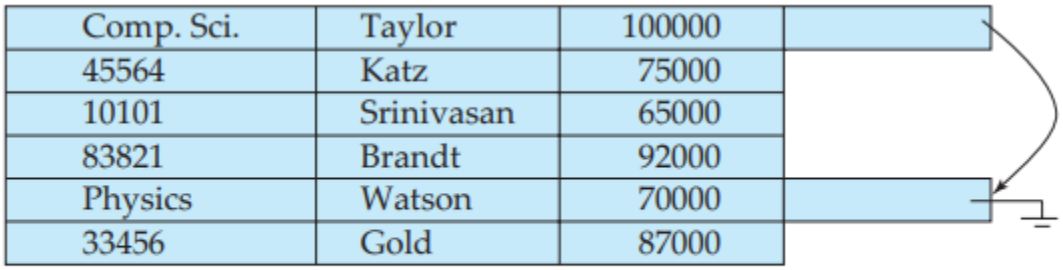


Figure 10.15 Multitable clustering file structure with pointer chains.

When multitable clustering is to be used depends on the types of queries that the database designer believes to be most frequent. Careful use of multitable clustering can produce significant performance gains in query processing.

Data-Dictionary Storage

So far, we have considered only the representation of the relations themselves. A relational database system needs to maintain data about the relations, such as the schema of the relations. In general, such “data about data” is referred to as **metadata**.

Relational schemas and other metadata about relations are stored in a structure called the data dictionary or system catalog. Among the types of information that the system must store are these:

- Names of the relations.
- Names of the attributes of each relation.
- Domains and lengths of attributes.
- Names of views defined on the database, and definitions of those views.
- Integrity constraints (for example, key constraints).

In addition, many systems keep the following data on users of the system:

- Names of authorized users.
- Authorization and accounting information about users.
- Passwords or other information used to authenticate users.

Further, the database may store statistical and descriptive data about the relations, such as:

- Number of tuples in each relation.
- Method of storage for each relation (for example, clustered or nonclustered).

The data dictionary may also note the storage organization (sequential, hash, or heap) of relations, and the location where each relation is stored:

- If relations are stored in operating system files, the dictionary would note the names of the file (or files) containing each relation.
- If the database stores all relations in a single file, the dictionary may note the blocks containing records of each relation in a data structure such as a linked list.

It also stores information about each index on each of the relations:

- Name of the index.
- Name of the relation being indexed.
- Attributes on which the index is defined.
- Type of index formed.

All this metadata information constitutes, in effect, a miniature database. Some database systems store such metadata by using special-purpose data structures and

code. It is generally preferable to store the data about the database as relations in the database itself. By using database relations to store system metadata, we simplify the overall structure of the system and harness the full power of the database for fast access to system data.

The exact choice of how to represent system metadata by relations must be made by the system designers. One possible representation, with primary keys underlined, is shown in Figure 10.16. In this representation, the attribute index attributes of the relation Index metadata is assumed to contain a list of one or more attributes, which can be represented by a character string such as “dept_name, building”. The Index metadata relation is thus not in first normal form; it can be normalized, but the above representation is likely to be more efficient to access. The data dictionary is often stored in a nonnormalized form to achieve fast access.

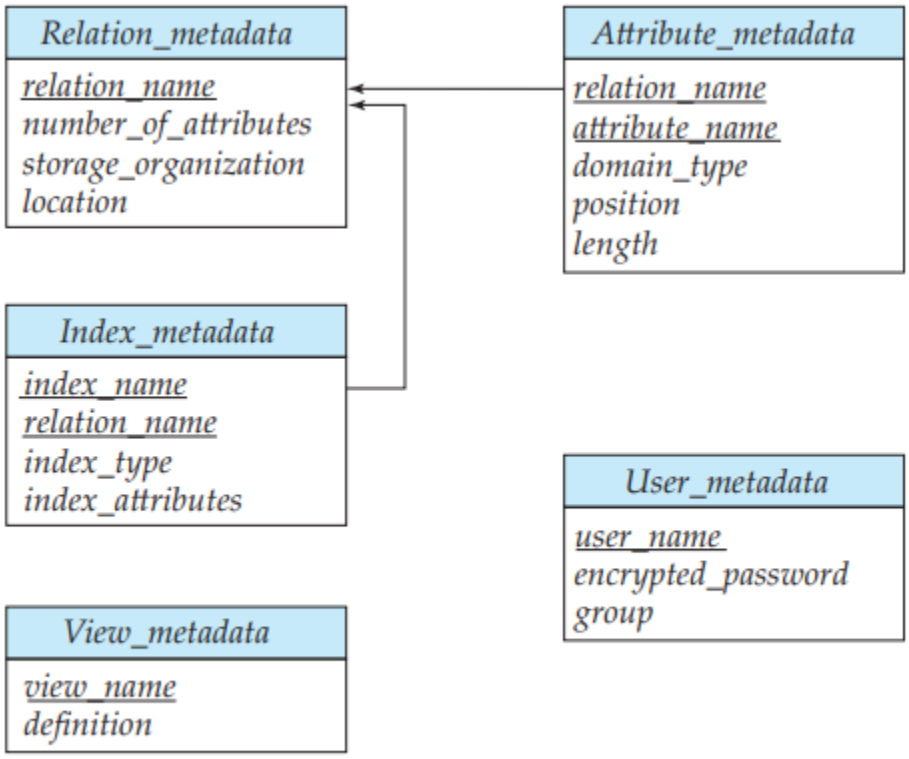


Figure 10.16 Relational schema representing system metadata.

Whenever the database system needs to retrieve records from a relation, it must first consult the Relation metadata relation to find the location and storage organization of the relation, and then fetch records using this information. However, the storage organization and location of the Relation metadata relation itself must be recorded elsewhere (for example, in the database code itself, or in a fixed location in the database), since we need this information to find the contents of Relation metadata.